

Chapter 7 - Complex Numbers

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#Notation

A complex number is denoted by $z = a + ib = (a, b)$ where $a = \text{Re}(z)$ and $b = \text{Im}(z)$.

#Product of Complex Numbers

The product of complex numbers $z = (a, b)$ and $w = (c, d)$, denoted by zw , is defined by
 $(a, b) \cdot (c, d) = (ac - bd, ad + bc)$

#The Imaginary Unit and Its powers

$i = \sqrt{-1}$	$i^5 = i$	$i^9 = i$	...	$i^{4n-3} = i$
$i^2 = -1$	$i^6 = -1$	$i^{10} = -1$	...	$i^{4n-2} = -1$
$i^3 = -i$	$i^7 = -i$	$i^{11} = -i$	...	$i^{4n-1} = -i$
$i^4 = 1$	$i^8 = 1$	$i^{12} = 1$	...	$i^{4n} = 1$

Note:

1. $i^{-1} = \frac{1}{i} = -i$
2. $\sqrt{-a}\sqrt{-b} \neq \sqrt{(-a)(-b)}$

#Complex Plane and Argand Diagram

The rectangular plane in which the X-axis and Y-axis respectively represents the real part and imaginary part of a complex number is called the complex plane.

The geometric diagram formed by plotting the complex number in the complex plane is called the argand diagram.

#Conjugate of a Complex Number

The conjugate of a complex number $z = a + ib$ is $\bar{z} = a - ib$

Properties of Conjugates

1. $z + \bar{z} = 2 \operatorname{Re}(z)$ and $z - \bar{z} = 2i \operatorname{Im}(z)$
2. $\overline{z + w} = \bar{z} + \bar{w}$
3. $\overline{zw} = \bar{z} \cdot \bar{w}$
4. $\overline{\bar{z}} = z$
5. $z\bar{z} = |z|^2$

#(Absolute Value/Modulus) of a Complex Number

The (absolute value) or (modulus) of a complex number $z = a + ib$ is $|z| = \sqrt{a^2 + b^2}$

Properties of (Absolute Values/Moduli)

1. $|z| = |\bar{z}|$
2. $|z + w| \leq |z| + |w|$ (**The Triangle Inequality**)
3. $|zw| = |z| \cdot |w|$
4. $\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$
5. $|z|^2 = z\bar{z}$
6. $\operatorname{Re}(z) \leq |z|$ and $\operatorname{Im}(z) \leq |z|$

#Square Roots of a Complex Number

If $x + iy = \sqrt{a + ib}$ then

$$x^2 = \frac{\sqrt{a^2 + b^2} + a}{2} \text{ and } y^2 = \frac{\sqrt{a^2 + b^2} - a}{2}$$

Note: Above definition does not exactly give all the square roots, the exact definition must include $\operatorname{sgn}(b)$ in the roots, so we have:

$$\text{If } z = a + ib \text{ then } \sqrt{a + ib} = \pm[x + iy \operatorname{sgn}(b)] \text{ where } x^2 = \frac{\sqrt{a^2 + b^2} + a}{2} \text{ and } y^2 = \frac{\sqrt{a^2 + b^2} - a}{2}$$